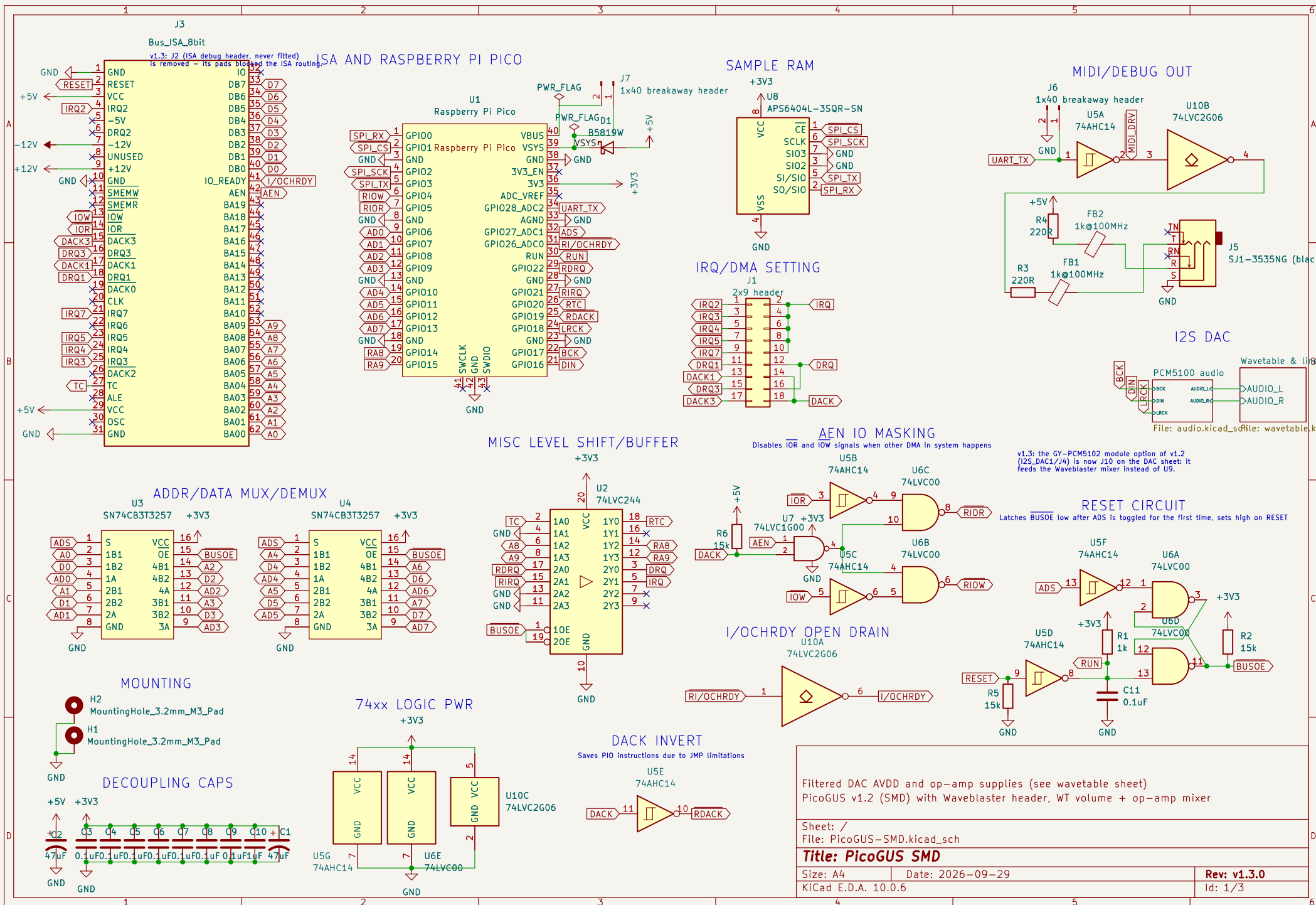




ISA AND RASPBERRY PI PICO

SAMPLE RAM

MIDI/DEBUG OUT

IRQ/DMA SETTING

I2S DAC

MISC LEVEL SHIFT/BUFFER

AEN IO MASKING

RESET CIRCUIT

I/OCHRDY OPEN DRAIN

DACK INVERT

ADDR/DATA MUX/DEMUX

MOUNTING

74xx LOGIC PWR

DECOUPLING CAPS

Filtered DAC AVDD and op-amp supplies (see wavetable sheet)
PicoGUS v1.2 (SMD) with Waveblaster header, WT volume + op-amp mixer

Sheet: /
File: PicoGUS-SMD.kicad_sch

Title: PicoGUS SMD

Size: A4
KiCad E.D.A. 10.0.6

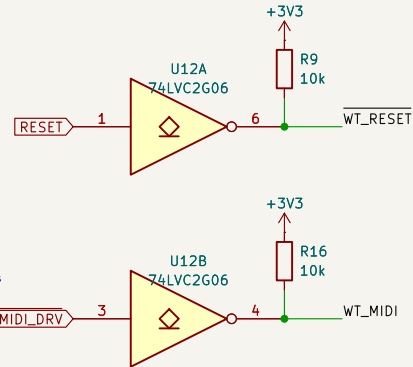
Date: 2026-09-29

Rev: v1.3.0

Id: 1/3

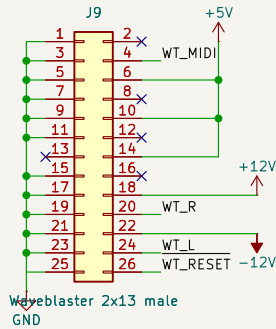
WT CONTROL (U12 = 74LVC2G06, open-drain inverters @ +3V3)

Both outputs are open drain with 10k pull-ups to +3V3, so the daughterboard never sees 5 V on /RESET or MIDI (some DBs have 3.3 V inputs).
/RESET: low while ISA RESET is high. MIDI: same data as the MIDI-out jack (MPU-401 UART).



WAVEBLASTER HEADER

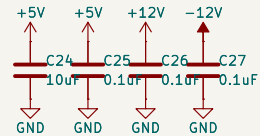
2x13 male header, 2.54 mm. Daughterboard (female socket) sits over the card; footprint has the standard standoff holes.



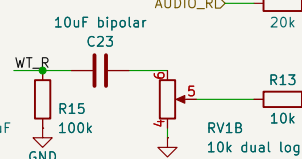
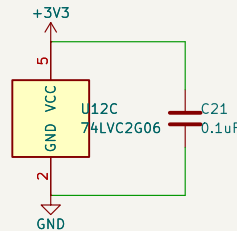
DGND 1-11, AGND 15-25:
both to the ground plane

Waveblaster 2x13 male
GND

WT SUPPLY DECOUPLING (place at J9)

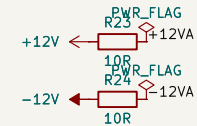


U12 POWER (+3V3) + DECOUPLING (place at U12)

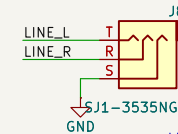
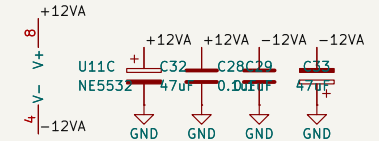


WAVETABLE VOLUME + MIXER

RV1 (dual 10k log) sets the WT level; pins 3/6 = CW end (signal), 1/4 = CCW end (GND).
U11 is an inverting summer: PCM 20k in / 20k fb (-unity, -0.2 dB), WT 10k in (x2 = +6 dB at RV1 max; DB outputs are -1-1.2 Vrms vs 2.1 Vrms PCM). The WT knob does not change the PCM level. C22/C23 (bipolar) block DC from the DB. R14/R15 keep them discharged when no DB is fitted. Output: 100R, DC-coupled like v1.2.



U11 SUPPLY: ISA +/-12 V via R23/R24 (10R) + C32/C33 (47u), C2



LINE OUT (green)

Derived from the hand386 ISA adapter mixer and the PicoGUS 2.0 WT header
Waveblaster (wavetable) daughterboard header, analog WT volume and op-amp mixer

Sheet: /Wavetable & line out/
File: wavetable.kicad_sch

Title: PicoGUS SMD – Waveblaster header & line-out mixer

Size: A4 Date: 2026-09-28

KiCad E.D.A. 10.0.6

Rev: v1.3.0

Id: 3/3